

Chia-Chien (Aaron) Hung

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EDUCATION

Georgia Institute of Technology

Atlanta, GA

Bachelor of Science in Computer Science, Concentration in Intelligence and Modeling/Simulation August 2023 - Present

Bachelor of Science in Biochemistry, Pre-Health Concentration

- GPA: 4.0/4.0 (Faculty Honors)
- Relevant Coursework: Machine Learning, Artificial Intelligence, Design and Analysis of Algorithms, Computer Organization and Programming, Systems and Networks, Differential Equations, Data Structures and Algorithms

EXPERIENCE

Undergraduate Teaching Assistant

Atlanta, GA

CS 2200: Systems and Networks

August 2025 - Present

- Taught processor architecture, operating systems, and networking in weekly labs for 30+ students.
- Designed office hours queue system serving 600+ students, improving TA response efficiency during peak demand.
- Developing automated grading tools projected to cut grading workload by ~50% and improving turnaround time.

Student Researcher

Atlanta, GA

Integrative Systems Biology Lab, Georgia Tech

December 2024 - Present

- Developed computational models of dynein-driven metastasis using PDE solvers and stochastic simulations to mimic experimental conditions, allowing low-cost computational simulations to be done before costly experiments.
- Applied vectorization and parallelism with Eigen and OpenMP, achieving 8× acceleration on Slurm HPC clusters.
- Optimized simulation routines by reducing algorithmic complexity while adhering to experimental conditions.

Student Researcher

Atlanta, GA

SIPLab, Georgia Tech

May 2024 - Present

- Developed a model predictive controller in C/C++, achieving 25× runtime improvement through optimization.
- Built Python + Snakemake pipelines for large-scale spiking neural network simulations and parameter optimization.
- Implemented and tested deep reinforcement learning controllers for real-time control in nonlinear systems.

Software Engineering Lead

Atlanta, GA

Molecular Evolution Core Lab, Georgia Tech

January 2024 - Present

- Lead large team building 5 modular Lab Information Management Systems, improving lab service maintainability 8×.
- Reduced client data transfer time 10× by developing integrated high-speed SFTP solution.
- Established agile sprints and CI/CD pipelines to ensure reliable deployment and collaborative development.

PROJECTS

Pipelined Processor

Atlanta, GA

CS 2200: Systems and Networks

April 2025

- Designed a custom pipelined processor with microcoded control logic for a MIPS-like architecture.
- Achieved 5.45× speedup via hazard detection, forwarding, and branch prediction.
- Optimized pipeline for throughput, significantly reducing stall cycles while minimizing hardware requirements.

Pin Preview – Augmented Reality Circuit Assistant

Atlanta, GA

HackGT 12

September 2025

- Won 2nd in the Impiricus track for building next-generation technology-enabled HCP engagement solutions.
- Built a full-stack web application that allows healthcare providers to search PubMed and in-house EMRs for similar patient cases, returning auto-summarized insights in real time.
- Designed and implemented text-embedding pipelines with Sentence Transformers and generative AI to parse user input, generate adaptive multi-tier search queries, and rank relevant patients/case studies.

SKILLS AND COMPETENCIES

Competitive Programming: Codeforces Rating: 1797

Programming Languages: C/C++, Python, Java, JavaScript/TypeScript, HTML/CSS, React, LaTeX, Assembly

Development Tools: Git, OpenCV, CMake, Neovim, Android Studio, Arduino, Linux, Agile, CI/CD

Specialties: High-performance computing, numerical simulation, backend architecture, microarchitecture